

Varak Tanashian

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Education

University of Toronto

*Honours Bachelor of Science in Computer Science and Statistics,
Minor in Mathematics*

Toronto, ON

Expected Graduation: 2028

Projects

Taction: Bayesian Model for Asset Prediction | *Python, PyTorch, FastAPI, Parquet* 🔄

- Built a data pipeline with pagination and asynchronous feature engineering for financial time-series analysis.
- Implemented Amazon's **DeepAR (LSTM)** architecture to forecast asset prices with quantified uncertainty.
- Developed an **XGBoost classification layer** for interest rate volatility based on macroeconomic indicators.
- Leveraged DS3 to engineer user-friendly visualizations of **probability distributions** to facilitate risk assessment.
- Deployed a production-ready API using **FastAPI** on Vercel, utilizing **Cloudflare** for DNS management.

Fundi: Pure Vanilla Java Finance App | *Java, SQL, JUnit, Clean Architecture* 🔄

- Collaborated in an Agile team of 5 to architect a financial management system following **SOLID** principles and **Clean Architecture**.
- Built a backend for purchase handling, budget validation, and basic trend prediction using a layered entity model.
- Integrated **Alpha Vantage**, **TwelveData**, **Finnhub**, and **ExchangeRate** for live equity and currency data.
- Managed user account and stock ownership persistence with **SQL**, including input validation and error handling.

Billboard-Artists: End-to-End Music Analytics | *FastAPI, TypeScript, D3.js* 🔄

- Worked in a team of 4 to build an automated ETL pipeline to scrape, normalize, and engineer features from Billboard Global 100 data.
- Developed a predictive engine using a **Graph Neural Network (GNN)** to predict Billboard Global 100 debut.
- Integrated a **FastAPI** backend with a **D3TypeScript** frontend as an accessible interface.

Précis: Speedy Chatbot Extension & API | *Python, PyTorch, HuggingFace, NLP* 🔄

- Acquired and denoised datasets to improve input signal quality.
- Fine-tuned the **Phi-4 Mini** and **Qwen-4:3b** for abstractive summarization of articles and transcripts.
- Engineered a **recursive chunking and merging pipeline** to bypass Transformer token limits.

Experience

Hackathon Competitor & Project Developer

June 2024 – Present

Multiple Events

Toronto, ON

- Mastered rapid prototyping through **Git**, **Docker**, and **Agentic AI** to deliver innovative solutions.
- Worked in fast-paced teams to design and build end-to-end software solutions under 36-hour constraints.
- Experience in DevOps practices to collaborate effectively in a team and win prizes.

Student Groups

Jan. 2025 – Present

UofTAI, UTMIST, and more

Toronto, ON

- Gained experience in workshops and project management.
- Learned about the internal workings of AI research and development.
- Became the lead developer for multiple clubs, building websites and applications, and organizing events.

Mathematics and Computer Science Tutor

Jan. 2023 – Dec. 2024

University / High School Private Tutoring

Toronto, ON

- Delivered instruction in **Data Structures, Algorithms, and Calculus**, both one-on-one and in group settings.
- Developed **Python-based visual aids** to illustrate computational logic and mathematical applications.
- Improved student examination performance by a measured average of **18%**.

Technical Skills

Languages: Python, C++, C, TypeScript, Javascript, Java, Shell, Assembly, HTML, CSS

Frameworks: PyTorch, TensorFlow, HuggingFace, Keras

Libraries: NumPy, Pandas, UNIX, Bash, Matplotlib, React, D3.js, SciPy

DevOps: Git, Docker, Kubernetes, GitHub Actions (CI/CD), Cloudflare Workers, Data Pipelines

Tools: Claude, Cursor, RestAPI, FastAPI, Flask, PostgreSQL, JUnit, Pytest